

```
Access Lexicons
1 # Assignment 1, Page 4: Access Lexicons
2 from nltk.corpus import stopwords
3 from nltk.corpus import names
4 from nltk.corpus import cmudict
5 #print("\nStop words in English: ", stopwords.words('english'))
6
7 import nltk
8 inaugural_corpus_words = nltk.corpus.inaugural.words()
9
10 def stopwords_percentage(data):
11     english_stopwords = stopwords.words('english')
12     content = [element for element in data if element.lower() in english_stopwords]
13     return(100 * (len(content) / len(data)))
14
15 print("\nPercentage of Stop Words in the Inaugural Corpus: ", stopwords_percentage(inaugural_corpus_words))
16
17 male_names = names.words('male.txt')
18
19 male_names_startingwith_Za = [element for element in male_names if element.startswith('Za')]
20 print("\nMale Names Starting with 'Za' in the Names Lexicon Corpus : ", male_names_startingwith_Za)
21
22 female_names = names.words('female.txt')
23 female_names_endingwith_a = [element for element in female_names if element.endswith('a')]
24 print("\nPercentage of Female Names Ending with 'a' in the Names Lexicon Corpus: ", 100-len(female_names_endingwith_a)/len(female_names))
25
```

Access Lexicons

Definition: Lexicon

- A lexicon, often referred to as a lexical resource, is a collection of words and/or phrases, such as a set of vocabulary or a dictionary, marked with allied information, such as each given word's part of speech or definition.
- A lexicon is often considered a type of corpus because it represents text data.

There are several lexicons included in the NLTK library.

Access Lexicons

There are several lexicons contained in the NLTK library. Some commonly used lexicons are listed and described below.

- **The Stopwords Corpus:** Words like "me", "has", "also", and "to" are all examples of stop words. Stop words add little meaning to text data. This lexicon corpus contains stop words in English.
- **The Names Corpus:** Categorized by gender, this lexicon corpus contains over 8,000 first names. Female names are stored in the file called `female.txt` and male names are stored in the file called `male.txt`.
- **The CMU Pronouncing Dictionary Corpus:** Based on US English, this lexicon corpus contains the phonetic pronunciations of words. Each phonetic pronunciation is represented with a symbol based on the ARPabet.

Enter the code below to access each of the above-mentioned lexicons.

```
from nltk.corpus import stopwords
from nltk.corpus import names
from nltk.corpus import cmudict
```

Use stopwords

For this first exercise, we will work with the lexicon corpus called `stopwords`. Using the corpus reader function `words`, we can return a list of strings representing English stop words.

Enter the code below, then select the `TRY IT` button to run the file.

```
print("\nStop words in English: ", stopwords.words('english'))
```

TRY IT

Stop Words in English: ['I', 'me', 'my', 'myself', 'we', 'our', 'ours', 'ourselves', 'you', 'you're', 'you've', 'you'll', 'you'd', 'your', 'yours', 'yourself', 'yourselves', 'he', 'him', 'his', 'himself', 'she', 'she's', 'her', 'hers', 'herself', 'it', 'it's', 'its', 'itself', 'they', 'them', 'their', 'theirs', 'themselves', 'what', 'which', 'who', 'whom', 'this', 'that', 'that'll', 'these', 'those', 'am', 'is', 'are', 'was', 'were', 'be', 'been', 'being', 'have', 'has', 'had', 'having', 'do', 'does', 'did', 'doing', 'is', 'am', 'the', 'and', 'but', 'if', 'on', 'because', 'as', 'until', 'while', 'of', 'at', 'by', 'for', 'with', 'about', 'against', 'between', 'into', 'through', 'during', 'before', 'after', 'above', 'below', 'to', 'from', 'up', 'down', 'in', 'out', 'on', 'off', 'over', 'under', 'again', 'further', 'then', 'once', 'here', 'there', 'when', 'where', 'why', 'how', 'all', 'any', 'both', 'each', 'few', 'more', 'most', 'other', 'some', 'such', 'no', 'nor', 'not', 'only', 'own', 'same', 'so', 'than', 'too', 'very', 's', 't', 'can', 'will', 'just', 'don', 'don't', 'should', 'should've', 'now', 'd', 'll', 'm', 'o', 're', 've', 'y', 'ain', 'aren', 'aren't', 'couldn', 'couldn't', 'didn', 'didn't', 'doesn', 'doesn't', 'hadn', 'hadn't', 'hasn', 'hasn't', 'haven', 'haven't', 'isn', 'isn't', 'ma', 'mightn', 'mightn't', 'mustn', 'mustn't', 'needn', 'needn't', 'shan', 'shan't', 'shouldn', 'shouldn't', 'wasn', 'wasn't', 'weren', 'weren't', 'won', 'won't', 'wouldn', 'wouldn't']

Your output should display as shown below.

Stop Words in English: ['I', 'me', 'my', 'myself', 'we', 'our', 'ours', 'ourselves', 'you', 'you're', 'you've', 'you'll', 'you'd', 'your', 'yours', 'yourself', 'yourselves', 'he', 'him', 'his', 'himself', 'she', 'she's', 'her', 'hers', 'herself', 'it', 'it's', 'its', 'itself', 'they', 'them', 'their', 'theirs', 'themselves', 'what', 'which', 'who', 'whom', 'this', 'that', 'that'll', 'these', 'those', 'am', 'is', 'are', 'was', 'were', 'be', 'been', 'being', 'have', 'has', 'had', 'having', 'do', 'does', 'did', 'doing', 'is', 'am', 'the', 'and', 'but', 'if', 'on', 'because', 'as', 'until', 'while', 'of', 'at', 'by', 'for', 'with', 'about', 'against', 'between', 'into', 'through', 'during', 'before', 'after', 'above', 'below', 'to', 'from', 'up', 'down', 'in', 'out', 'on', 'off', 'over', 'under', 'again', 'further', 'then', 'once', 'here', 'there', 'when', 'where', 'why', 'how', 'all', 'any', 'both', 'each', 'few', 'more', 'most', 'other', 'some', 'such', 'no', 'nor', 'not', 'only', 'own', 'same', 'so', 'than', 'too', 'very', 's', 't', 'can', 'will', 'just', 'don', 'don't', 'should', 'should've', 'now', 'd', 'll', 'm', 'o', 're', 've', 'y', 'ain', 'aren', 'aren't', 'couldn', 'couldn't', 'didn', 'didn't', 'doesn', 'doesn't', 'hadn', 'hadn't', 'hasn', 'hasn't', 'haven', 'haven't', 'isn', 'isn't', 'ma', 'mightn', 'mightn't', 'mustn', 'mustn't', 'needn', 'needn't', 'shan', 'shan't', 'shouldn', 'shouldn't', 'wasn', 'wasn't', 'weren', 'weren't', 'won', 'won't', 'wouldn', 'wouldn't']

Try this variation:

With the use of the `stopwords` lexicon corpus, we can find the percentage of how often English stop words were used in Inaugural Addresses stored in the `inaugural` corpus.

Enter the code below, then select the `TRY IT` button to run the file.

```
import nltk
inaugural_corpus_words = nltk.corpus.inaugural.words()

def stopwords_percentage(data):
    english_stopwords = stopwords.words('english')
    content = [element for element in data if element.lower() in english_stopwords]
    return(100 * (len(content) / len(data)))

print("\nPercentage of Stop Words in the Inaugural Corpus: ", stopwords_percentage(inaugural_corpus_words))
```

TRY IT

Percentage of Stop Words in the Inaugural Corpus: 47.59746502638962%

Your output should display as shown below.

Percentage of Stop Words in the Inaugural Corpus: 47.59746502638962%

Use names

For this second exercise, we will work with the lexicon corpus called `names`. Using the corpus reader function `words`, we can create a variable called `male_names` to represent all male names contained in this corpus. Then, we can analyze this data, such as by returning all male names that start with a "Za".

Enter the code below, then select the `TRY IT` button to run the file.

```
s.words('male.txt')

length_Za = [element for element in male_names if element.startswith('Za')]
ma Starting with 'Za' in the Names Lexicon Corpus : (male_names_startingwith_Za)
```

TRY IT

Percentage of Stop Words in the Inaugural Corpus: 47.59746502638962%

Male Names Starting with 'Za' in the Names Lexicon Corpus : ['Zach', 'Zacharia', 'Zachariah', 'Zacharias', 'Zacharie', 'Zachary', 'Zacharia', 'Zachery', 'Zack', 'Zackariah', 'Zak', 'Zalman', 'Zane', 'Zared', 'Zary']

Your output should display as shown below.

Male Names Starting with 'Za' in the Names Lexicon Corpus : ['Zach', 'Zacharia', 'Zachariah', 'Zacharias', 'Zacharie', 'Zachary', 'Zacharia', 'Zachery', 'Zack', 'Zackariah', 'Zak', 'Zalman', 'Zane', 'Zared', 'Zary']

Try this variation:

Write the code to produce the following expected output, then select the `TRY IT` button to run the file and check your work.

The expected output is shown below.

```
ma Ending with 'a' in the Names Lexicon Corpus : 35.45298948161388

female_names_endingwith_a = [element for element in female_names if element.endswith('a')]
female_percentage = 100 * (len(female_names_endingwith_a) / len(female_names))
print("\nPercentage of Female Names Ending with 'a' in the Names Lexicon Corpus: ", female_percentage)
```

TRY IT

Percentage of Stop Words in the Inaugural Corpus: 47.59746502638962%

Male Names Starting with 'Za' in the Names Lexicon Corpus : ['Zach', 'Zacharia', 'Zachariah', 'Zacharias', 'Zacharie', 'Zachary', 'Zacharia', 'Zachery', 'Zack', 'Zackariah', 'Zak', 'Zalman', 'Zane', 'Zared', 'Zary']

Percentage of Female Names Ending with 'a' in the Names Lexicon Corpus: 35.45298948161388

▼ Solution

One possible solution is shown below. Store all of the female names in a variable. Use the `index()` method to find all of the female names that end with `a`. Calculate the percentage of female names ending with `a` as compared to the entire list of female names. Print the result.

```
female_names = names.words('female.txt')

female_names_endingwith_a = [element for element in female_names if element.endswith('a')]
female_percentage = 100 * (len(female_names_endingwith_a) / len(female_names))
print("\nPercentage of Female Names Ending with 'a' in the Names Lexicon Corpus: ", female_percentage)
```

Reading Question

Which definition **best** describes a lexicon?

- ☐ A lexicon, or lexical resource, is a set of vocabulary, a list of words, or a dictionary.
- ☒ A lexicon, or a lexical resource, represents text along with each text component's associated data.
- ☐ A lexicon, or lexical resource, is used to identify the part of speech or definition of each component in a specified text.
- ☐ A lexicon, or lexical resource, is a copy of a corpus used to analyze the original corpus' data.

The correct answer is:

A lexicon, or a lexical resource, represents text along with each text component's associated data.

Incorrect: A lexicon, or lexical resource, is a copy of a corpus used to analyze the original corpus' data. This is a false statement, as a lexicon is not a copy of a corpus. However, a lexicon is often considered a type of corpus because it contains a body of text.

Incorrect: A lexicon, or lexical resource, is a set of vocabulary, a list of words, or a dictionary. This is a false statement, as a lexicon is not simply a set of vocabulary, a list of words, or a dictionary. Rather, a lexicon can be any of these items, allied with associated data.

Incorrect: A lexicon, or lexical resource, is used to identify the part of speech or definition of each component in a specified text. This is a false statement, as lexicons are not only used to identify the parts of speech or definitions of components in a specified text. Lexicons can be designed to possess this information about a set of text.